

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 2.7 SURFACE AREA AND VOLUME OF SOLIDS: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: Why does the same amount of metal sheet make containers that hold very different amounts of grain, and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction? Use what you observed and calculated in Lessons 1 to 8 to write a full explanation. Answer every section in the space under the prompt. Show your working for every calculation, use $\pi = 22/7$, and include units (cm², cm³, m², m³, litres). Refer to Mzee Kiplagat's posho mill in Eldoret, Wanjiru the jua kali metalsmith, and your own class evidence. Remember: 1 litre = 1,000 cm³ and 1 m³ = 1,000 litres.

Part 1: The puzzle and the box

Mzee Kiplagat's posho mill in Eldoret has two closed containers made by Wanjiru: a cylindrical drum and a square-based box. Wanjiru says she used exactly 30,800 cm² of iron sheet for each. (a) Explain how you can use a net to find the total surface area of a prism. (b) The box measures 50 cm x 50 cm x 129 cm. Calculate its total surface area and say whether it agrees with Wanjiru's figure. (c) Explain why knowing the surface area alone does not tell Mzee Kiplagat how much maize a container holds.

(a) Open the prism out into its net, find the area of every face, and add them. This gives TSA = 2 x base area + perimeter of base x length (height). (b) Base area = $50 \times 50 = 2,500 \text{ cm}^2$, so two bases = $5,000 \text{ cm}^2$. The perimeter of the base is $4 \times 50 = 200 \text{ cm}$, so the four side faces = $200 \times 129 = 25,800 \text{ cm}^2$. TSA = $5,000 + 25,800 = 30,800 \text{ cm}^2$. This agrees with Wanjiru's figure, so the box really used 30,800 cm² of sheet. (c) Surface area measures the material covering a solid, while volume measures the space inside. They are different quantities, and capacity depends on volume. So two containers can have the same surface area and still hold different amounts of maize.

Part 2: Testing the drum

The drum has radius 35 cm and height 105 cm. (a) Decompose the closed drum into its parts and calculate its total surface

(a) The drum is two circles plus one curved wall. $2 \times \pi \times r^2 = 2 \times 22/7 \times 35 \times 35 = 7,700 \text{ cm}^2$. Curved surface = $2 \times \pi \times r \times h = 2 \times 22/7 \times 35 \times 105 = 23,100 \text{ cm}^2$. TSA = $7,700 + 23,100 = 30,800 \text{ cm}^2$. Yes, this is the same as the box and agrees with Wanjiru's figure. (b) Drum: $V = \pi \times r^2 \times h = 22/7 \times 35 \times 35 \times 105 = 404,250 \text{ cm}^3 = 404.25 \text{ litres}$. Box: $V = \text{base area} \times \text{height} = 2,500 \times 129 = 322,500 \text{ cm}^3 = 322.5 \text{ litres}$. (c) The drum holds more: $404.25 - 322.5 = 81.75 \text{ litres}$ more maize. This matches Mzee Kiplagat's

area. Does the method reproduce Wanjiru's figure? (b) Calculate the volume of the drum and the volume of the box in cm³, then convert each to litres. (c) Which container holds more maize, and by how many litres? Does this match what Mzee Kiplagat saw?	observation that the drum holds noticeably more grain even though both used 30,800 cm² of sheet.
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Part 3: Explaining why shape matters

Use your evidence from Parts 1 and 2 to explain the main puzzle. (a) Complete this idea in your own words: 'Equal surface area does not mean equal volume because...'. (b) Using the cross-section of each container, explain why the drum's shape encloses more space than the box's shape for the same amount of sheet. (c) Of the solids you studied, which shape would hold the most for a given surface area? Explain why a sphere is not always the best choice for a real grain store.	(a) Equal surface area does not mean equal volume because surface area measures only the material covering a solid, while volume measures the space inside. The two are different quantities, so shape, not just the amount of sheet, decides capacity. Our evidence: both containers used 30,800 cm², but the drum holds 404,250 cm³ and the box 322,500 cm³. (b) The drum has a circular cross-section of area $22\frac{7}{8} \times 35 \times 35 = 3,850$ cm², and the box has a square cross-section of area $50 \times 50 = 2,500$ cm². The drum's cross-section is 1,350 cm² larger. The box is actually taller (129 cm against 105 cm), but its extra 24 cm of height does not make up for its smaller base, so the drum's greater base area outweighs the height difference and the drum holds more maize (404,250 cm³ against 322,500 cm³). In general, a circle encloses more area than a square for the same perimeter, which is why round shapes tend to give more capacity for the same sheet. (c) A sphere is the best of all, since it encloses the most space for the least covering. In practice a sphere is hard to make from sheet metal, hard to stack and store, and costly, so artisans balance capacity against stacking and cost.
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Part 4: Surface area question or volume question?

The Rift Valley cooperative wants a water tank made of a cylinder (radius 2.1 m, height 4 m, flat base) with a hemisphere on top, and a grain silo made of a cylinder (radius 2.1 m, height 6 m) with a cone roof. Mzee Kiplagat also has a pyramid roof (6 m x 6 m base, vertical height 4 m) on a small store. (a) For each of these jobs, say whether you need surface area	(a) (i) Surface area, because paint covers the outside. (ii) Volume, because capacity is the space inside. (iii) Surface area, because it is the covering material. (iv) Volume. (b) The slant height comes from Pythagoras: half the base is 3 m, so $l^2 = 3 \times 3 + 4 \times 4 = 25$ and $l = 5$ m. Roof area (four triangular faces only) = $4 \times \frac{1}{2} \times 6 \times 5 = 60$ m². Volume = $\frac{1}{3} \times \text{base area} \times \text{vertical height} = \frac{1}{3} \times 36 \times 4 = 48$ m³. We use the vertical height for volume and the slant height for the triangular faces. (c) Decompose the solid into familiar parts and add only the outside faces. The joining circle is hidden inside the solid, so it is left out of the surface area. This is why the tank has SA = 94.38 m² and the silo has TSA = 116.16 m². For volume we add the volumes of the parts, giving 74.844 m³ (74,844 litres) for the tank and 96.096 m³ for the silo.
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or volume: (i) painting the tank, (ii) finding how much water the tank holds, (iii) finding how much iron sheet covers the pyramid roof (no floor), (iv) finding the space enclosed by the pyramid roof. (b) Calculate the roof sheet needed in (iii) and the volume in (iv). (c) When you find the surface area of a composite solid such as the tank, what must you do about the circle where two parts join? Why?	
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Part 5: Using mathematics to design	
Answer the KICD inquiry questions using your own project or the class evidence. (a) How do we determine the surface area and volume of solids? Give at least three different solids and the formula or method you used for each. (b) How are surface area and volume applied in real life in Kenya? Describe one design for a named Kenyan community need (for example a grain silo, water tank or sufuria), say which calculation you would do for sheet or paint and which for capacity, and name one career that uses this mathematics. (c) Finally, write a short final explanation (3 to 5 sentences) answering the driving question.	(a) Prism: $TSA = 2 \times \text{base area} + \text{perimeter} \times \text{length}$, and $V = \text{base area} \times \text{height}$. Pyramid: $TSA = \text{base area} + \text{areas of the triangular faces (using slant height } l)$, and $V = \frac{1}{3} \times \text{base area} \times \text{vertical height}$. Cone: curved $SA = \pi \times r \times l$, $TSA = \pi \times r(r + l)$, and $V = \frac{1}{3} \times \pi \times r^2 \times h$. Frustum: curved $SA = \pi(R + r)l$, and $V = \text{big cone minus small cone}$. Sphere: $SA = 4 \times \pi \times r^2$ and $V = \frac{4}{3} \times \pi \times r^3$. Composite solids: split into known solids, add only the outside faces for SA, and add the volumes for V. (b) Example: a grain silo for a Rift Valley cooperative, a cylinder with a cone roof. The surface area tells the metal fabricator how much iron sheet to buy and how much paint is needed, and the volume tells how many litres or m³ of maize it stores. A metal fabricator, civil engineer, architect, packaging engineer or quantity surveyor uses both calculations to give the needed capacity for the least sheet and cost. (c) The drum and the box in Mzee Kiplagat's posho mill both used 30,800 cm² of sheet, but the drum holds 404,250 cm³ (404.25 litres) and the box 322,500 cm³ (322.5 litres), so the drum holds 81.75 litres more. Surface area measures the material covering a solid and volume measures the space inside, so they are different quantities. The drum's circular cross-section (3,850 cm²) is larger than the box's square cross-section (2,500 cm²), and this outweighs the box's extra height, so shape decides capacity; in general a circle encloses more area than a square for the same perimeter and a sphere is best of all. Artisans like Wanjiru use mathematics to choose shapes and dimensions that give the most capacity for the least sheet, balanced against stacking and cost.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Calculating surface area and	Uses the correct method for each solid	Uses mostly correct methods and gets	Confuses formulae, mixes slant and

volume correctly	(prism, cylinder, pyramid, cone, sphere, composite), uses slant height and vertical height correctly, gets all values right (30,800 cm ² ; 404,250 cm ³ ; 322,500 cm ³) with correct units and working.	most values right, with one or two small slips in arithmetic, units or choice of height.	vertical height, or omits working and units, so the values are mostly incorrect.
Distinguishing surface area from volume	States clearly that surface area is the material covering a solid and volume is the space inside, and correctly classifies each task as a surface area or volume question with reasons.	Distinguishes the two ideas correctly but gives limited reasons or misclassifies one task.	Treats surface area and volume as the same thing or cannot say which one a task needs.
Explaining the phenomenon with evidence	Explains why the drum holds 81.75 litres more than the box using the shared 30,800 cm ² , both volumes and the cross-section areas (3,850 cm ² against 2,500 cm ²), noting that the drum's larger base outweighs the box's extra height, and links to the sphere as the most efficient shape.	Explains that shape decides capacity and quotes the main numbers, but the reasoning about cross-sections is incomplete.	States that the drum holds more without evidence, or suggests that equal surface area means equal volume.
Handling composite solids and real-life application	Leaves out hidden joined faces in surface area, adds the volumes of the parts, and describes a realistic Kenyan design, explaining which calculation gives sheet and which gives capacity, plus a relevant career.	Handles composite solids mostly correctly and gives a reasonable Kenyan application with some detail missing.	Counts hidden faces or gives no clear real-life application.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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